Design of novel Ciprofloxacin derivatives as potential antimicrobial agents based on molecular docking studies and ADMET prediction

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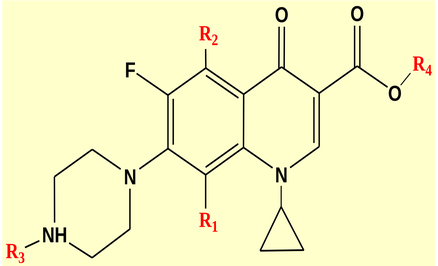


Scheme Molecular structures of **Cip** derivatives

	R1	R2	R3	R4
Cip	H	H	H	-
1-cip	F	H	H	-
2-cip	H	F	H	-
3-cip	F	F	H	-
4-cip	CF_3	H	H	-
5-cip	H	CF_3	H	-
6-cip	CF_3	F	H	-
7-cip	NH_2	H	H	-
8-cip	OCH_4	H	H	-
9-cip	H	CF_3	CH_2COOH	-
10-cip	H	H	CH_2COOH	-
11-cip	H	H	H	CH_3
12-cip	H	H	H	C_2H_5
13-cip	F	H	H	CH_3
14-cip	F	H	H	C_2H_5
15-cip	H	CF_3	H	CH_3
16-cip	H	CF_3	H	C_2H_5

Introduction

- **Ciprofloxacin**, is one of a new generation of **fluorinated quinolones**, her first mechanism of action is inhibition of bacterial DNA gyrase.
- ***Bacillus Mycobacterium tuberculosis*** is the causative agent of tuberculosis. It is one of the top 10 causes of death and represents one of the major challenges facing drug discovery.
- The replacement of one or more hydrogen atoms by fluorine atoms is used to improve metabolic stability, bioavailability and protein-ligand interactions. When CF_3 is combined with various organic molecules and the addition of groups, alkyl, alkoxy and hydroxyl play an important role in improving the performance of drugs.
- **In silico** methods are increasingly used in the design strategies of new molecules for therapeutic purposes "**Drug design**".
- The ciprofloxacin derivatives have predicted theoretically in order to design 16 novel molecules to investigate their antimicrobial activity.
- **Molecular docking** have been performed to investigate the binding energy and interactions with the amino acid residues of receptor proteins.
- **Pharmacokinetic properties** (**ADMET**) were estimated for all candidates.

Computational methodology

✓ We have performed the geometry optimization of structure, molecular electrostatic potential (MEP) and then NBO analysis were estimated to examining all possible interactions between the 'occupied' and the 'none-occupied' NBO orbitals. using **G 09W program** at **B3LYP/6-311G(d,p)**.

✓ Analysis of protein-ligand interactions with **molecular docking** using **auto dock vina** and discover studio 2020 for visualizations.

✓ The prediction of the ADMET parameters of all derivatives with the **ADMET-LAB platform**.

Molecular electrostatic potential (MEP) of CIP and 9-CIP

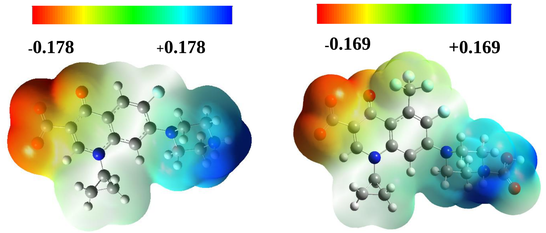


Fig. 2 Molecular electrostatic potential maps of Cip and 9-cip.

NBO analysis of CIP and 9-CIP

Interactions Types	$E^{(2)}$ (kcal/mol)	
	Cipro	9-cip
$\pi_{C11-C16} \rightarrow \pi^*_{C17-C18}$	49.42	-
$\pi_{C11-C16} \rightarrow \pi^*_{C19-C24}$	38.53	-
$\pi_{C17-C18} \rightarrow \pi^*_{C19-C24}$	51.53	42.73
$\pi_{C19-C24} \rightarrow \pi^*_{C11-C16}$	34.79	-
$\pi_{C19-C24} \rightarrow \pi^*_{C17-C18}$	26.33	31.14
$n_{3\ F1} \rightarrow \pi^*_{C17-C18}$	16.08	1.36
$n_{1\ N6} \rightarrow \pi^*_{C19-C24}$	13.91	14.13
$n_{1\ N6} \rightarrow \sigma^*_{C20-H33}$	12.86	12.77
$n_{1\ N6} \rightarrow \sigma^*_{C23-H37}$	11.65	11.41
$n_{1\ C16} \rightarrow \pi^*_{C17-C18}$	-	126.78
$n_{3\ F40} \rightarrow \pi^*_{C19-C24}$	-	-
$n_{3\ F42} \rightarrow \sigma^*_{C41-F43}$	-	11.78
$n_{3\ F44} \rightarrow \sigma^*_{C41-F43}$	-	13.82
$n_{2\ O46} \rightarrow \pi^*_{C39-O45}$	-	66.30

Docking molecular score and interaction of best Cip derivative with β-lactamase (PDB:3N7W)

	3N7W
Cip	-7.5
1-cip	-7.6
2-cip	-7.7
3-cip	-7.9
4-cip	-7.7
5-cip	-8.0
6-cip	-7.5
7-cip	-7.0
8-cip	-7.3
9-cip	-8.7
10-cip	-8.0
11-cip	-7.1
12-cip	-7.2
13-cip	-7.4
14-cip	-7.3
15-cip	-7.8
16-cip	-8.0

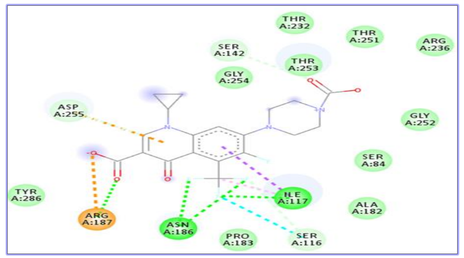
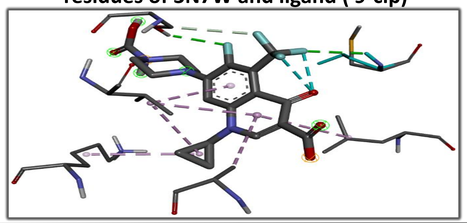


Fig.1: The interactions between active site residues of 3N7W and ligand (9-cip)



Pharmacokinetic proprieties ADMET of ciprofloxacin and its derivatives

	CACO-2	HIA	PPB	BBB	CYP 2D6 inhibitor	CYP 2D6 substrat	hERG	LD_{50}
Cip	-5.12	Y	41.91	N	N	N	N	421.90
1-cip	-5.10	Y	40.17	N	N	N	N	379.52
2-cip	-5.10	Y	40.87	Y	N	N	N	394.68
3-cip	-5.01	Y	41.62	Y	N	N	N	345.98
4-cip	-4.93	Y	65.24	N	N	N	N	327.60
5-cip	-4.93	Y	65.36	Y	N	N	N	424.95
6-cip	-4.88	Y	60.75	Y	N	N	N	351.95
7-cip	-5.16	Y	36.54	N	N	N	N	400.82
8-cip	-5.33	Y	35.14	N	N	N	N	377.16
9-cip	-5.19	Y	53.88	N	N	N	N	309.27
10-cip	-5.37	Y	49.87	N	N	N	N	356.70
11-cip	-5.13	Y	42.48	N	N	N	N	866.07
12-cip	-5.10	Y	61.88	N	N	N	N	915.78
13-cip	-5.07	Y	45.50	N	N	N	N	1154.90
14-cip	-5.06	Y	60.77	N	N	N	N	1081.30
15-cip	-4.94	Y	68.30	N	N	N	N	238.45
16-cip	-4.94	Y	74.86	Y	N	N	N	205.52

Conclusion

- We note that the compound **9-cip** pointed out its **great inhibition** behavior with β-lactamase.
- **ADMET** results of compound **9-cip**:
  - Show an **excellent absorptive property** and demonstrates that it is not potentially carcinogenic.
  - According to values of PPB % he has a **moderate bioavailability** for receptor target.
  - We note that he **exhibit good metabolism**, has a **lower risk of developing** heart problems and manifest **medium to low acute** toxicity in rat models.
- **NBO analyses** showed that the delocalization energies were very high and therefore the compound 9-cip is stable enough to show desired medicinal properties also that could be the reason for his activity.

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